10.22 Collinear Logarithms in φ^3 Theory in Six Dimensions

LecNote15 pp. 1–3 (Srednicki §26) sends $m \to 0$, finds $\ln m$ in the $2 \to 2$ rate, and cures it with a detector that cannot separate two particles closer than an angle δ; p. 10 checks the cure in $\overline{\text{MS}}$. Students ask:

- "Where do p. 3's $\frac{1}{3}$ and $\delta^2 s$ come from?"
- "p. 1 says 'collinear or soft'. Where is the soft factor?"
- "On p. 10 the residue's $\ln m$ cancels against the emission. Luck? And which theorem is p. 3's 'KLT'?"
- "What is new at $m = 0$ on p. 2?"

In short: a line of energy $E = \sqrt{s}/2$ becomes an unresolved pair with probability $\frac{\alpha}{12}\ln(\delta^2 s/m^2) + O(m^0)$, and four lines give $\frac{\alpha}{3}$ (for the two incoming ones, under an assumption). Soft emission is finite in $d = 6$. Each leg's splitting cancels the $\ln m$ of that leg's residue R: both count the same two-particle spectral weight, whose m-dependence sits at threshold, and at $m = 0$ the threshold sits on the one-particle pole. The theorem is Kinoshita–Lee–Nauenberg (KLN). Notation is that of §10.15 (φ^3, $d = 6$, metric $(-,+,+,+)$, $\alpha = g^2/(4\pi)^3$). Page numbers without a LecNote refer to LecNote15, counted as PDF pages (circled number -8).

The splitting probability

The $2 \to 3$ amplitude has a propagator $1/(K^2 + m^2)$, $K = k_1 + k_2$, that blows up at $m = 0$ when two outgoing momenta are parallel (p. 1). A detector blind to angles below δ books such events as $2 \to 2$. There the $2 \to 3$ amplitude is the $2 \to 2$ one with outgoing momentum K, times $ig \cdot \frac{1}{i}\tilde{\Delta}(K)$, so an outgoing line of momentum k multiplies the rate by $1 + P$, with P p. 3's margin factor:

$$P = \frac{1}{2}\int \widetilde{dk}_1\,\widetilde{dk}_2\,(2\pi)^5\,2\omega\,\delta^5(\mathbf{k} - \mathbf{k}_1 - \mathbf{k}_2)\,\frac{g^2}{(K^2 + m^2)^2}\,\Theta(\delta - \theta_{12}). \tag{10.22.1}$$

Here $\omega = (\mathbf{k}^2 + m^2)^{1/2}$, θ_{12} is the opening angle, $\frac{1}{2}$ is for identical particles, and $E_1 + E_2 - \omega$ is neglected.
For $E \equiv \omega \gg m$ and $\theta_{12} \ll 1$, write $z = E_1/E$ and $k_\perp = z(1-z)E\theta_{12}$, the relative transverse momentum. Then $-(K^2 + m^2) = [k_\perp^2 + (1 - z + z^2)m^2]/z(1-z)$, and with $d^5k_1 = |\mathbf{k}_1|^4 d|\mathbf{k}_1|\,2\pi^2\sin^3\theta_1\,d\theta_1$, $\theta_1 = (1-z)\theta_{12}$,

$$P = \frac{\alpha}{2}\int_0^1 dz\,z(1-z)\int_0^{z^2(1-z)^2E^2\delta^2}\frac{k_\perp^2\,dk_\perp^2}{\left[k_\perp^2 + (1 - z + z^2)m^2\right]^2}. \tag{10.22.2}$$

Four transverse dimensions, $d^4k_\perp \propto k_\perp^3\,dk_\perp$, against the squared propagator, $k_\perp^{-4}$: the inner integral is $\int dk_\perp^2/k_\perp^2$ from m^2 to $E^2\delta^2$, the collinear logarithm. For $m \ll E\delta$ it is $\ln[z^2(1-z)^2E^2\delta^2/(1-z+z^2)m^2]-1$, and the z integrals give

$$P = \frac{\alpha}{12}\ln\frac{E^2\delta^2}{m^2} + \Big(\frac{1}{9} - \frac{\pi\sqrt{3}}{12}\Big)\alpha, \qquad \frac{m}{E} \ll \delta \ll 1. \tag{10.22.3}$$

Here $\frac{1}{12} = \frac{1}{2}\int_0^1 z(1-z)\,dz$, and the constant is $-0.342\,\alpha$. At $E = \sqrt{s}/2$ this is $\frac{\alpha}{12}\ln(\delta^2 s/m^2) + O(m^0)$ per line, and four lines give p. 3's $\frac{\alpha}{3}$, inside its window $m^2/|\mathbf{k}|^2 \ll \delta^2 \ll 1$. Exact kinematics at $E = 10^4 m$ agree with (10.22.3) to $3\times10^{-4}\alpha$ for $\delta = 0.1$ to 0.02; at $E = 10^3 m$, where $m/E\delta$ reaches 0.1, the slope $dP/d\ln\delta^2$ drops to 0.065.

Incoming lines. p. 3 counts all four lines. For the two incoming ones Srednicki asserts the same factor (§26, just before (26.14)). It holds when the initial state, too, is known only to within δ, so that an unresolved pair entering the collision (the time reverse of the splitting, the same integral (10.22.1)) counts as the single particle. That is KLN's sum over degenerate initial states (below). In a sharply prepared beam, a particle that sheds a collinear one before colliding enters with only part of its energy, so its emission is not a factor $1+P$ on $|\mathcal{T}|^2$; that case is not treated here.

Why only collinear

At $m = 0$ give one particle of the pair a small energy ω_q at a fixed angle θ: $K^2 = -2E\omega_q(1-\cos\theta)$, the emission factor goes as ω_q^{-2}, the phase space $d^5q/2\omega_q$ as $\omega_q^3\,d\omega_q$, and the soft region gives $\int \omega_q\,d\omega_q$: finite. At $m = 0$ the integrand of (10.22.2) is $\alpha\,z(1-z)\,dz\,d\theta_{12}/\theta_{12}$, logarithmic in the angle only. A soft photon in QED has the same ω_q^{-2}, but in four dimensions its phase space is $\omega_q\,d\omega_q$, and $\int d\omega_q/\omega_q$ diverges (in d dimensions: $\int \omega_q^{d-5}d\omega_q$). The general result is the KLN theorem ("KLT" on p. 3 is a slip): transition probabilities summed over the initial and final states degenerate in energy, soft and collinear ones included, are free of mass singularities order by order in perturbation theory. For soft photons in QED the Bloch–Nordsieck theorem suffices: rates that include the photons below the detector's energy threshold are infrared finite.

Where $\ln m$ lives: the cancellation leg by leg

On shell (p. 3) the cancellation is partial: $|\mathcal{T}|^2$ carries $-\frac{11}{6}\alpha\ln(s/m^2)$ (§10.21), emission adds $\frac{1}{3}\alpha\ln(\delta^2 s/m^2)$, and $-\frac{3}{2}\alpha\ln(s/m^2)$ is left. Its m is the on-shell reference point: m and the residue were fixed at $k^2 = -m^2$ and g at zero momentum, points that become branch points at $m = 0$ (p. 4). In $\overline{\text{MS}}$ the self-energy, vertex and box are finite at $m = 0$:

$\Pi_{\overline{\rm MS}} \to \frac{\alpha}{12}k^2[\ln(k^2/\mu^2) - \frac{8}{3}]$, the vertex with two legs on shell keeps only $\int dF_3 \ln(x_1x_2) = -3$ besides its $\ln(s/\mu^2)$, and the box tends to its massless form (§10.21). So in $\mathcal{T} = R^2\mathcal{T}_0[1 - \frac{11}{12}\alpha\ln(s/\mu^2) + O(m^0)]$ (p. 10; $R^{1/2}$ per leg, §10.11) all of $\ln m$ is in the residue of p. 9,

$$R^{-1} = 1 - \Pi'_{\overline{\rm MS}}(-m^2) = 1 + \frac{\alpha}{12}\Big[\ln\frac{\mu^2}{m^2} + c''\Big], \qquad c'' = \frac{17 - 3\pi\sqrt{3}}{3}. \tag{10.22.4}$$

Each leg puts a factor R into $|\mathcal{T}|^2$ and $1 + P$ into the observed rate, and by (10.22.3)

$$R\,(1 + P) = 1 + \alpha\Big[\frac{1}{12}\ln\frac{E^2\delta^2}{\mu^2} - \frac{13}{36}\Big] + O(\alpha^2). \tag{10.22.5}$$

The $\pi\sqrt{3}$'s cancel too: both are $\frac{1}{2}\int_0^1 dz\, z(1-z)\ln(1 - z + z^2)$, from the mass term of (10.22.2) and from LecNote12's $D_0 = (1 - x + x^2)m^2$ in R. With exact kinematics, $\delta = 0.05$ and $\mu = E$, $[R(1 + P) - 1]/\alpha = -0.8575$, -0.8603, -0.8604 at $m/E = 10^{-3}$, 10^{-4}, 10^{-5}; (10.22.5) gives -0.8604. Four legs with $E^2 = s/4$ give LecNote15 p. 10's result,

$$|\mathcal{T}_{\rm obs}|^2 = |\mathcal{T}_0|^2\Big[1 - \alpha\Big(\frac{3}{2}\ln\frac{s}{\mu^2} + \frac{1}{3}\ln\frac{1}{\delta^2}\Big) + O(m^0)\Big], \tag{10.22.6}$$

with no $\ln m$ (its red check). The $\ln(s/\mu^2)$ that is left is the subject of §10.24.

The spectral view

For $m > 0$ the one-loop spectral density $\rho = \frac{\alpha}{12}s\beta^3/(s - m^2)^2$ (10.15.12) starts at $4m^2$, above the pole at m^2 (LecNote15 p. 2, upper sketch). At $m = 0$ it is $\alpha/12s$ down to $s = 0$: two collinear massless particles have $K^2 = 0$, like one, and the continuum reaches the pole (lower sketch). That is what is new on p. 2, and R and P count this same weight. Since $\Pi'(-m^2)$ depends on the scheme only through its $-A$ and vanishes on shell, $R^{-1} - 1 = A_{\overline{\rm MS}} - A_{\rm OS}$, and $-A_{\rm OS} = \int \rho\, ds$ in $d = 6 - \epsilon$ by (10.15.8) and (10.15.14). Inserting $1 = \int dK^0\,\delta(K^0 - E_1 - E_2)$ turns (10.22.1) into the pair phase space of (10.15.11). Hence

$$R^{-1} = 1 + \Big[\int_{4m^2}^{\infty} ds\,\rho(s)\Big]_{\overline{\rm MS}}, \qquad P = \int ds\,\rho(s)\,\frac{\omega}{K^0}\,f_\delta(s), \tag{10.22.7}$$

where $[\ ]_{\overline{\rm MS}}$ drops the pole $\alpha/6\epsilon$, $K^0 = (\mathbf{k}^2 + s)^{1/2}$, and $f_\delta(s)$ is the fraction of pairs of mass^2 s with $\theta_{12} < \delta$. So $R(1 + P) - 1 = -\big[\int ds\,\rho\,(1 - f_\delta\,\omega/K^0)\big]_{\overline{\rm MS}}$. For $s \ll E^2\delta^2$ nearly every pair is unresolved and $\omega/K^0 \approx 1$, so the integrand vanishes where ρ depends on m, at $s \sim m^2$. At $m = 0$ the pole cannot be told from the collinear pairs and R alone diverges (p. 4); a rate summed over pairs up to $s \sim E^2\delta^2$ never asks for the separation. Hence "infrared" on p. 1: $\ln m$ comes from pairs near threshold, not from large loop momenta.